

AI LAB WEEK-1

TIC TAC TOE

```
board = ['_'] * 9
```

```
def show():
```

```
    print(board[0], board[1], board[2])
```

```
    print(board[3], board[4], board[5])
```

```
    print(board[6], board[7], board[8])
```

```
def won(p):
```

```
    lines = [
```

```
        (0, 1, 2), (3, 4, 5), (6, 7, 8),
```

```
        (0, 3, 6), (1, 4, 7), (2, 5, 8),
```

```
        (0, 4, 8), (2, 4, 6)
```

```
    ]
```

```
    for a, b, c in lines:
```

```
        if board[a] == board[b] == board[c] == p:
```

```
            return True
```

```
    return False
```

```
turn = 'X'
```

```
for i in range(9):
```

```
    show()
```

```
    pos = int(input(f"{turn}'s move (0-8): "))
```

```
if 0 <= pos <= 8 and board[pos] == '_':
```

```
    board[pos] = turn
```

```
    if won(turn):
```

```
        show()
```

```
        print(turn, "wins!")
```

```
        break
```

```
    turn = 'O' if turn == 'X' else 'X'
```

```
else:
```

```
    print("Invalid move. Try again.")
```

```
    i -= 1
```

```
else:
```

```
    print("Draw!")
```

```
-- --
-- --
-- --
X's move (0-8): 0
X _ _
-- --
O's move (0-8): 1
X O _
-- --
X's move (0-8): 2
X O X
-- --
O's move (0-8): 6
X O X
-- --
O _ _
O _ _
X's move (0-8): 8
X O X
-- --
O _ X
O's move (0-8): 0
Invalid move. Try again.
X O X
-- --
O _ X
O's move (0-8): 5
X O X
-- --
O _ O
O _ X
X's move (0-8): 4
X O X
-- --
X O
O _ X
X wins!
```

VACUUM CLEANER AGENT

```
A_is_dirty = True
B_is_dirty = True
current_location = 'A'

print("Simplified Vacuum Cleaner Simulation")
print("-----")

for step in range(5):
    current_location_is_dirty = False

    if current_location == 'A':
        current_location_is_dirty = A_is_dirty
    else:
        current_location_is_dirty = B_is_dirty

    print(
        f"Step {step + 1}: "
        f"Location={current_location}, "
        f"Status={'Dirty' if current_location_is_dirty else 'Clean'}"
    )

    if current_location_is_dirty:
        print("  Action: Suck")

        if current_location == 'A':
            A_is_dirty = False
        else:
            B_is_dirty = False

    elif not A_is_dirty and not B_is_dirty:
        print("  Action: Stop - goal reached")
        break

    else:
        print("  Action: Move")
        current_location = 'B' if current_location == 'A' else 'A'

    print(
        f"  Current State: "
        f"A={'Dirty' if A_is_dirty else 'Clean'}, "
        f"B={'Dirty' if B_is_dirty else 'Clean'}"
    )
    print("-----")

if not A_is_dirty and not B_is_dirty:
    print("Simulation successful: Both locations are clean!")
else:
    print("Simulation ended after 5 steps. Goal may not have been reached.")
```

Simplified Vacuum Cleaner Simulation

Step 1: Location=A, Status=Dirty

Action: Suck

Current State: A=Clean, B=Dirty

Step 2: Location=A, Status=Clean

Action: Move

Current State: A=Clean, B=Dirty

Step 3: Location=B, Status=Dirty

Action: Suck

Current State: A=Clean, B=Clean

Step 4: Location=B, Status=Clean

Action: Stop – goal reached

Simulation successful: Both locations are clean!